

Osmotic Pressure Regulating Drainage Valve

Executive Technology Brief

Domain: Osmotic Membrane Transport + Membrane Mechanics

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Fixed-pressure valves cannot adapt to changing CSF conditions

Mechanism: Semipermeable membrane with osmotic reservoir provides self-regulating drainage based on combined hydrostatic and osmotic pressure

Architecture: CSF shunt valve regulated by osmotic pressure differential across semipermeable membrane. Osmotic agent concentration on one side of membrane modulates effective drainage pressure based on CSF protein concentration or implanted osmotic reservoir.

Governing Model: Semipermeable membrane flux governed by Kedem-Katchalsky equations. Net water flux depends on hydraulic pressure differential and osmotic pressure differential, scaled by membrane permeability L_p and reflection coefficient σ .

Key Equations:

$$p_i = i * C * R * T \text{ [osmotic pressure]}$$

$$J_v = L_p * (dP - \sigma * d\pi) \text{ [Starling/Kedem-Katchalsky]}$$

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$$Q_{\text{drainage}} = J_v * A_{\text{membrane}} \text{ [volumetric drainage rate]}$$

$$p_i = i * C * R * T \text{ [van 't Hoff osmotic pressure]}$$

Kill Condition: Membrane fouling rate in CSF reduces L_p by >50% within 30 days (insufficient for chronic implant)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.